

Task-Time Morphology–Motion Co-Design under Reconfiguration Feasibility

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Extended abstract, IROS 2026 Workshop on Learning-based Robot Co-design

Abstract—Self-reconfigurable robots can change embodiment during a task, which creates a design decision that fixed-morphology robots never face: *where can the robot physically transform?* We formulate task-time co-design over route, morphology, reconfiguration site, and transition feasibility, and ask whether modeling the physical feasibility of the transformation itself changes which design is selected and how reliably it executes. On a simulated six-pod self-mobile robot with two qualified morphologies, we compare route-first adaptation, geometry-coupled search, and feasibility-coupled search across three diagnostic environments (10 trials each, 90 missions total) run through the complete Gazebo, Nav2, localization, reconfiguration, and evaluator stack. At obstructed sites, feasibility-coupled search completed the transformation without contact in 20/20 missions, versus 17/20 for geometry-coupled search and 19/20 for route-first adaptation; the unobstructed control showed an 83% (25/30) baseline completion rate across planners from disturbance alone, indicating the gap between feasibility-coupled and geometry-coupled search is real but modest at this sample size (Fisher’s exact, $p \approx 0.12$). An earlier diagnostic pass also surfaced a post-transition verification motion, an in-place yaw, that was outside the original transition model and caused contact even at feasibility-accepted sites; we extended the model to cover this motion, which we believe accounts for the improvement to 10/10 in the affected condition. Preliminary as they remain, the results indicate that where a robot can physically change embodiment, and what motions it performs while doing so, should be first-class variables in task-time co-design.

I. MOTIVATION AND POSITIONING

Robot co-design has largely optimized a body *before* deployment [1]–[3]. Self-reconfigurable modular robots instead choose embodiment *during* a task: prior work selects configurations for task/environment requirements [4], [5], plans paths across discrete morphologies [6], and plans module motions or connectivity changes between configurations [7], [8]. Task-level planners commonly treat a transformation as a switch between two admissible configurations, while reconfiguration planners take the target and its location as given. The coupling between the two: two configurations can each be collision-free at a location while the module motions that connect them do not fall between these layers.

We contribute a task-time co-design formulation that adds transition feasibility as a first-class variable alongside route, morphology sequence, and reconfiguration site. Learned morphology search [2], [3] is complementary: it expands the design space, whereas we ask whether and where an available embodiment change can physically occur within a mission.

II. PLATFORM AND PLANNERS

The robot is a sensing core with six independently mobile differential-drive pods in Gazebo/ROS 2 (Fig. 1a). Two morphologies are qualified end to end: `compact_diff` and a longer, narrower `narrow_tandem` for constrained passages. Reconfiguration is executed, not teleported: pods detach, drive along declared relocation paths, align using connector observations, and latch; execution inhibits assembled motion until a valid commit. The planner searches over (x, y, θ, m) , with morphology-specific traversal costs and a pre-action failure probability on transition edges.

Three planners share mechanics, controllers, map, and disturbance realization. **Route-first** fixes a spatial route, then inserts morphology changes along it. **Geometry-coupled** searches route and morphology jointly, admitting a transition when a planar clearance disk and the target footprint are free. **Feasibility-coupled** additionally samples every relocating module’s 3D collision boxes along its trajectory against the environment, stationary core, and support/latch conditions, rejecting the edge with a structured reason. Route-first $\rightarrow$ geometry-coupled isolates coupling route and embodiment; geometry-coupled $\rightarrow$ feasibility-coupled isolates physical transition feasibility.

III. DIAGNOSTIC STUDY

One passage environment, generated by shared code, has three variants. The doorway admits only `narrow_tandem`. *Blocked-A* places a low post beside the attractive staging site so that before/after footprints remain admissible but the outward relocation of one pod intersects it; *Blocked-B* mirrors the post to intercept a different pod; *Neutral* removes it. The post lies below the lidar plane and outside the assembled robot’s corridor, so it constrains only the transformation. Geometry-coupled search staged at $x = 2.15$ m in both blocked variants; feasibility-coupled search rejected the corresponding edges for module–environment collision and staged at $x = 1.85$ m (Fig. 1b,c). All three planners staged at $x = 2.15$ m in *Neutral* with no rejections.

An initial single-trial pass (3 variants $\times$ 3 planners) through the full Gazebo, Nav2, localization, reconfiguration, and evaluator stack surfaced a failure mode outside the original transition model: an automatic post-transition motion check performs an in-place yaw of the assembled body, which in *Blocked-A* swung a pod into the post independent of relocation-path feasibility. We extended the feasibility model to cover this verification motion, then replicated the study

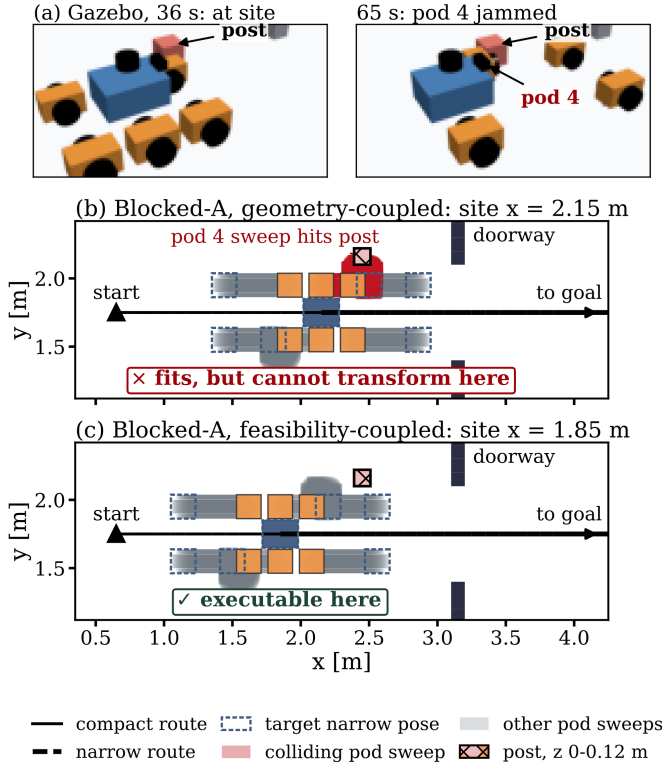


Fig. 1. (a) Gazebo renders of the Blocked-A mission: 36s, staged at the site with the post visible; 65s, pod 4 in sustained contact, producing the `RECOVERY_REQUIRED` jam of Section IV; executed-mission captures. (b)–(c) planner/scenario geometry only. (b) Geometry-coupled: the post (light-red) lies outside the robot’s corridor but inside pod 4’s outward relocation sweep (light-red) at the preferred site (×), consistent with (a). (c) Feasibility-coupled search rejects that edge, staging 0.3 m earlier (✓).

at $n = 10$ full missions per (environment, planner) cell—90 missions total. Results below (Section IV) reflect the corrected model, reported explicitly because it is why feasibility-coupled outcomes in Blocked-A differ from the single-trial figures in Fig. 1.

IV. PRELIMINARY RESULTS

Single-trial pilot. In Blocked-A, geometry-coupled planning staged where both morphologies fit (Fig. 1b) but the relocating pod contacted the post for 40s and the transformation did not complete (`RECOVERY_REQUIRED`); feasibility-coupled planning rejected that edge, staged 0.3 m earlier, and completed with no contact. Blocked-B repeated the pattern; route-first matched feasibility-coupled’s site and outcome in both cases; Neutral showed no difference across planners. Under the original transition model, however, neither transformed Blocked-A mission completed end-to-end: an unmodeled post-transition yaw swung a pod into the post regardless of relocation-path feasibility, motivating the correction in Section III.

Replication at $n = 10$. Table I reports mission completion without contact across 10 independently disturbed trials per cell, using the corrected model. At obstructed sites, feasibility-coupled search completed every mission without

TABLE I
MISSION COMPLETION WITHOUT CONTACT, $n = 10$ PER CELL.

Environment	Route-first	Geometry-coupled	Feasibility-coupled
Blocked-A	9/10	8/10	10/10
Blocked-B	10/10	9/10	10/10
Neutral	8/10	8/10	9/10
Obstructed (A+B)	19/20 (95%)	17/20 (85%)	20/20 (100%)
All conditions	27/30 (90%)	25/30 (83%)	29/30 (97%)

contact (20/20), versus 17/20 for geometry-coupled and 19/20 for route-first. The direction matches the pilot, but the margin is smaller: geometry-coupled search still succeeded on most obstructed trials, since trial-to-trial pose/friction variation shifts the pod’s sweep and does not always intersect the post. A Fisher’s exact test on feasibility- vs. geometry-coupled outcomes at obstructed sites (20/20 vs. 17/20) gives $p \approx 0.12$ (one-sided)—directionally consistent but not statistically established at this sample size.

The Neutral condition’s 80–90% completion rate (8/10, 8/10, 9/10), despite no obstruction, indicates a ~ 10 –20% baseline failure rate from disturbance variance alone. The raw obstructed-site gap (100% vs. 85%) should be read against this noise floor rather than a 100% ceiling – the true obstruction-attributable effect is smaller than the raw percentages suggest, though still favorable to feasibility-coupled search. Planning time remained ~ 23 –52 s for coupled planners vs. ~ 1 s for route-first; clearance held at ~ 4 –5 cm as in the pilot.

V. DISCUSSION AND LIMITATIONS

The replication clarifies rather than simplifies the pilot result: geometry-coupled search’s failure at obstructed sites is real but probabilistic, depending on trial-to-trial variation in the relocating pod’s swept trajectory, which the geometry-only check never modeled. Feasibility-coupled search’s advantage held up at $n = 10$ but did not reach significance in this sample; we regard the mechanism, checking 3D swept collision along the actual relocation trajectory, now including the post-transition verification motion, as the more defensible claim than the specific completion-rate numbers, and flag $n = 20$ –30 per cell as a target for a full paper. The results here reflect a transition model revised after the pilot to cover the post-transition yaw; we consider that revision itself a finding-every motion performed at a site, not only relocation, belongs in the feasibility model.

More broadly, a site can accommodate both the pre- and post-transition robot and still be unable to host the transformation between them, so site and transition executability should be first-class co-design variables: extending beyond relocation to every motion performed at the site. These remain diagnostic cases: two morphologies, simulation only, idealized connectors, static obstacles as a known map, future work couples generative morphology proposals with transition feasibility.

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